Madeleine Grunde-McLaughlin

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Education

2021-present | **Ph.D. Candidate at the University of Washington,** *Seattle, WA.*
Paul G. Allen School of Computer Science & Engineering
Co-advised by Jeffrey Heer and Daniel Weld
Proposed thesis: Surfacing Alternatives to Support Human Oversight of Agentic AI Systems

2016-2021 | **B.A. at the University of Pennsylvania,** *Philadelphia, PA.*
Bachelor of Arts in Cognitive Science with Summa Cum Laude
Minors in Computer Science, French

2019 | **Community Auditing Program at Princeton University,** *Princeton, NJ.*
Audited Computer Vision, NLP, Advanced Graph Theory (not for credit)

2019 | **Study Abroad at Lyon Lumière II,** *Lyon, France.*
Courses in French including Neuroscience, Human Computer Interaction, and Memory

Manuscripts and Pre-prints

2027 | **Model Maps: A Documentation Framework for Tracing Foundation Model Ancestry and Understanding AI Entanglement**
Rachel Hong, ***Madeleine Grunde-McLaughlin****, Sarah Cen, Jevan Hutson, Jamie Morgenstern, Tadayoshi Kohno*
Under review at ACM FAccT 2027

2027 | **Agentic Path Maps: Exposing an Agent's Traversal Through the Garden of Forking Paths**
Madeleine Grunde-McLaughlin*, Hafsah Usman, Daniel Mitroshkov, Emily Reif, Nino Migineishvili, Lakshya Chauhan. Kevin Yu, Jeffrey Heer, Daniel S. Weld*
Under review at ACM CHI 2027

2027 | **Overseeing Agents Without Constant Oversight: Challenges and Opportunities**
Madeleine Grunde-McLaughlin*, Hussein Mozannar, Maya Murad, Jingya Chen, Saleema Amershi, Adam Fourney*
Under review at ACM IUI 2027

Publications

IUI 2026 | **RDoFlow: Automatically assessing under-specified statistical analyses in HCI**
Madeleine Grunde-McLaughlin*, Weixuan Liu, Ria Patil, Nino Migineishvili, Emily Reif, Ranjay Krishna, Daniel S. Weld, Jeffrey Heer*
ACM Conference on Intelligent User Interfaces

TOCHI 2026 | **Wildfire and Forest Management: Opportunities for HCI Research**
Nino Migineishvili, ***Madeleine Grunde-McLaughlin****, Emmanuel Azuh, Spencer Wood, Rene Just, Katharina Reinecke*
ACM Transactions on Computer-Human Interaction

TOCHI 2025 | **Designing LLM Chains by Adapting Techniques from Crowdsourcing Workflows**
***Madeleine Grunde-McLaughlin**, Michelle S Lam, Ranjay Krishna, Daniel S Weld, Jeffrey Heer*
ACM Transactions on Computer-Human Interaction

COLM 2025 | **Visual Representations inside the Language Model**
*Benlin Liu, Amita Kamath, **Madeleine Grunde-McLaughlin**, Winson Han, Ranjay Krishna*
Conference on Language Modeling

CHI 2024 | **How Do Data Analysts Respond to AI Assistance? A Wizard-of-Oz Study**
*Ken Gu, **Madeleine Grunde-McLaughlin**, Andrew M McNutt, Jeffrey Heer, Tim Althoff*
ACM Conference on Human Computer Interaction, 2024

CSCW 2023 | **Explanations can Reduce Overreliance on AI Systems during Decision-Making**
*Helena Vasconcelos, Matthew Jörke, **Madeleine Grunde-McLaughlin**, Ranjay Krishna, Tobias Gerstenberg, and Michael Bernstein*
ACM Conference on Computer-Supported Cooperative Work and Social Computing, 2023

CVPR 2022 | **AGQA-Decomp: Measuring Compositional Consistency for Video Question Answering**
*Mona Gandhi, Mustafa Öümer Gul, Eva Prakash, **Madeleine Grunde-McLaughlin**, Ranjay Krishna, Maneesh Agrawala*
IEEE conference on Computer Vision and Pattern Recognition, 2022

CVPR 2021 | **AGQA: A Benchmark for Compositional Spatio-Temporal Reasoning**
***Madeleine Grunde-McLaughlin**, Ranjay Krishna, Maneesh Agrawala*
IEEE conference on Computer Vision and Pattern Recognition, 2021

InfoVis 2020 | **Bayesian-Assisted Inference from Visualized Data**
*Yea-Seul Kim, Paula Kayongo, **Madeleine Grunde-McLaughlin**, Jessica Hullman*
IEEE Transactions of Visualization & Computer Graphics (Proceedings of InfoVis), 2020

Workshop papers and public manuscripts

ArXiv 2025 | **Magentic-ui: Towards human-in-the-loop agentic systems**
*Hussein Mozannar, Gagan Bansal, Cheng Tan, Adam Fourney, Victor Dibia, Jingya Chen, Jack Gerrits, Tyler Payne, Matheus Kunzler Maldaner, **Madeleine Grunde-McLaughlin**, Eric Zhu, Griffin Bassman, Jacob Alber, Peter Chang, Ricky Loynd, Friederike Niedtner, Ece Kamar, Maya Murad, Rafah Hosn, Saleema Amershi*
ArXiv, July 2025

NAACL 2025 | **Towards AI-assisted academic writing**
*Daniel J Liebling, Malcolm Kane, **Madeleine Grunde-Mclaughlin**, Ian Lang, Subhashini Venugopalan, Michael Brenner*
Nations of the Americas chapter of the ACL, Workshop on AI and Scientific Discovery

CSCW 2023 | **Semantic Navigator: Query Driven Active Learning for Historical Narrative Understanding**
*Eva Maxfield Brown, **Madeleine Grunde-McLaughlin**, Isabelle Pestovski, Lanyi Zhu, Nicholas Weber*
ACM Conference on Computer-Supported Cooperative Work and Social Computing, Community-Driven AI Workshop

CHI 2022 | **When Do XAI Methods Work? A Cost-Benefit Approach to Human-AI Collaboration**
*Helena Vasconcelos, Matthew Jörke, **Madeleine Grunde-McLaughlin**, Ranjay Krishna, Tobias Gerstenberg, and Michael Bernstein*
ACM Conference on Human Computer Interaction, TRAIT Workshop

ArXiv 2022 | **AGQA 2.0: An updated benchmark for compositional spatio-temporal reasoning**
***Madeleine Grunde-McLaughlin**, Ranjay Krishna, Maneesh Agrawala*
ArXiv, April 2022

Industry research internships

2025 | **Microsoft Research Intern** *Redmond, Washington*
AI Frontiers Group, mentored by Adam Fourney and Saleema Amershi

2023 | **Google Student Researcher** *Seattle, Washington*
Google Applied Sciences Group, mentored by Dan Liebling

Selected Awards and Honors

2026 | **IUI Best Paper Honorable Mention,** *top-5*
"RDoFlow: Automatically assessing under-specified statistical analyses in HCI"

2023 | **CSCW Best Paper Honorable Mention,** *top-5*
"Explanations can Reduce Overreliance on AI Systems during Decision-Making"

2021 | **Allen School Computer Science & Engineering Research Fellowship**
1-year fellowship from the University of Washington Allen School

2021 | **College Alumni Society Prize in Cognitive Science**
Awarded to the best Cognitive Science thesis at the University of Pennsylvania

2021 | **Phi Beta Kappa Honor Society**

Academic Service

2024-2026 | **DUB (Design Use Build) Student Coordinator**

2025-2026 | **Diverse Genders in Research Student Coordinator**

2025-2026 | **New Grad Mentor**

2025 | **Pre-Application Mentoring Service**

2024-2025 | **Inaugural DUB (Design Use Build) Community Day Session Chair Organizer**

2023-2025 | **New Grad Mentoring Program Organizer,**

2024 | **Visit Days Non-Standard Visitor Coordinator**

2023 | **Visit Days PCS Area HCI Lead Scheduler-in-Chief**

2023-2024 | **HCI Seminar Organizer**

2022-2023 | **Doctoral Colloquium Coordinator for DUB (Design Use Build)**

2022-2023 | **New Grad Mentor**

Paper Reviewing

2026-2027 | **ACM Conference on Intelligent User Interfaces (IUI)**

2024-2027 | **ACM Conference on Human Factors in Computing Systems (CHI)**

2022-2026 | **ACM Symposium on User Interface Software and Technology (UIST)**

Teaching experience

Fall 2025	**CSE 442 Data Visualization** Undergraduate course teaching assistant under Jeffrey Heer
Fall 2024	**CSE 512 Data Visualization** Graduate course teaching assistant under Jeffrey Heer
Spring 2024	**CSE 512 Data Visualization** Graduate course teaching assistant under Jeffrey Heer

Mentoring Experience gt

2026	**Hafsah Usman** Undergraduate at the UW School of Information
2026	**Lakshya Chauhan** Undergraduate in Computer Science & Engineering
2026	**Daniel Mitroshkov Shaffer** Undergraduate in Computer Science & Engineering
2026	**Kevin Yu** Undergraduate in Computer Science & Engineering
2024-2026	**Weixuan Liu** Undergraduate in Computer Science & Engineering, now a Master's student at CMU
2024-2025	**Ria Patil** Undergraduate in Computer Science & Engineering
2021-2022	**Mona Gandhi** Undergraduate at VJTI, now a PhD student at the University of Pennsylvania
2021-2022	**Mustafa Omer Gul** Masters at Stanford University, now a PhD student Cornell
2021-2022	**Eva Prakash** Undergraduate at Stanford University, now a PhD student at Stanford University